



THE KING'S SCHOOL

2007
Higher School Certificate
Trial Examination

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1-10
- All questions are of equal value

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Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.

Marks

- | | |
|--|---|
| (a) Calculate, correct to three significant figures, $12 \tan^2 1$ | 2 |
| (b) Simplify $\frac{x+1}{1+\frac{1}{x}}$ | 2 |
| (c) Solve $ 3x + 4 \leq 5$ | 2 |
| (d) Solve $3^x = 2$ correct to two decimal places. | 2 |
| (e) Differentiate $1 + \frac{1}{x}$ | 2 |
| (f) Find a primitive of $(2x + 9)^3$ | 2 |

End of Question 1

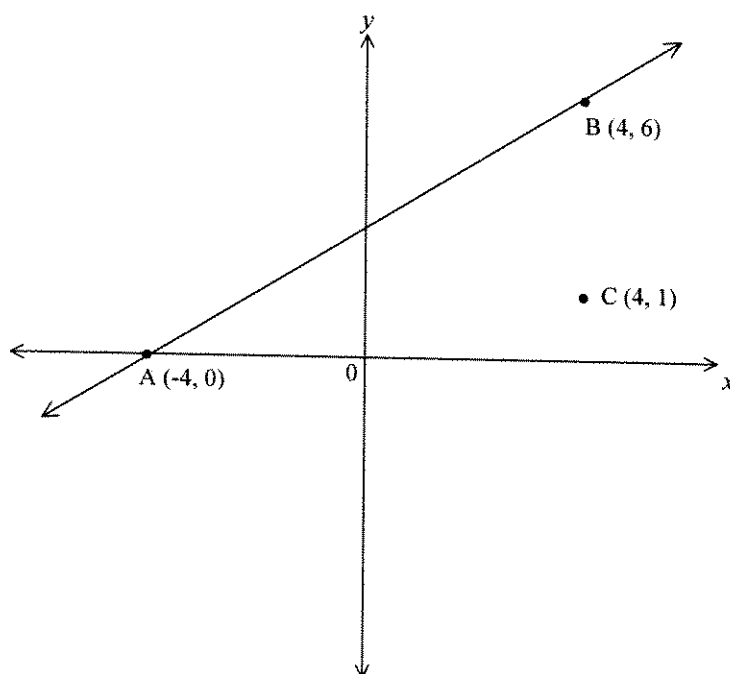
- (a) Solve simultaneously

$$y = 4x$$

$$y = 2x^2 - x + 2$$

3

- (b)



The diagram shows the points $A(-4, 0)$, $B(4, 6)$ and $C(4, 1)$. O is the origin.

- (i) Find the gradient of the line AB .

1

- (ii) Deduce that the equation of the line AB is $3x - 4y + 12 = 0$

2

- (iii) The perpendicular from $C(4, 1)$ meets line AB at D . i.e. $CD \perp AB$ at D . Find the length of CD .

2

- (iv) Find the length of DB .

1

- (c) Find the equation of the tangent to the curve $y = x^3 - 2x^2$ at the point $(-1, -3)$.

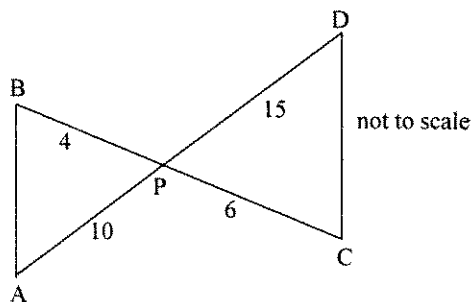
3

End of Question 2

- (a) Evaluate $\int_0^2 \frac{12x}{x^2 + 2} dx$ expressing your answer in simplest exact form.

3

(b)



In the diagram, BPC and APD are straight lines.

AP = 10, PD = 15, BP = 4 and PC = 6

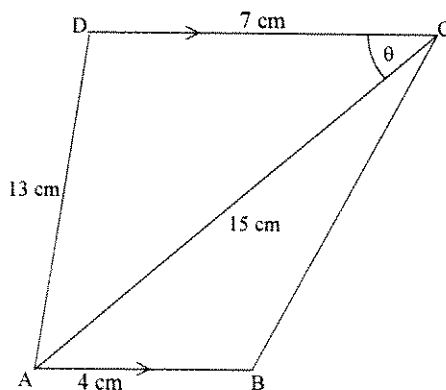
- (i) Prove $\triangle ABP$ is similar to $\triangle CDP$

2

- (ii) Deduce that $AB \parallel CD$

2

(c)



The diagram shows a trapezium ABCD where $AB \parallel DC$ and $AB = 4$ cm, $CD = 7$ cm, $DA = 13$ cm, $AC = 15$ cm.

Let $\angle DCA = \theta$

- (i) Find θ

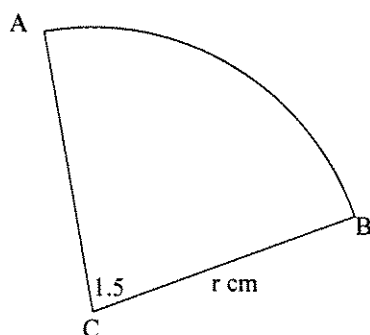
2

- (ii) Find the exact area of the trapezium.

3

End of Question 3

(a)



CAB is a sector of a circle with centre C and radius r cm. $\angle ACB = 1.5$ radians. The perimeter of the sector is 1.4 cm.

- | | |
|--|---|
| (i) Find r . | 2 |
| (ii) Find the area of the sector. | 1 |
| (iii) Find $\angle ACB$ correct to the nearest degree. | 1 |

(b) $2007 + 2000 + \dots$ is an arithmetic series.

- | | |
|--|---|
| (i) State the common difference. | 1 |
| (ii) Show working to decide whether 12 is a term in the series. | 2 |
| (iii) Find the maximum number of terms for which the sum of the series remains positive. | 2 |

- | | |
|---|---|
| (c) Solve the equation $79100 \times 1.002^{40} - M(1.002^{40} + 1.002^{39} + \dots + 1.002 + 1) = 0$ giving your value for M correct to the nearest integer. | 3 |
|---|---|

End of Question 4

- (a) For a particular curve $y = f(x)$, which passes through the point $(2, 0)$, we have $f'(x) = 3x(2 - x)$

(i) Determine the nature of the stationary points of the curve.

3

(ii) Prove that $f(-1) = 0$

3

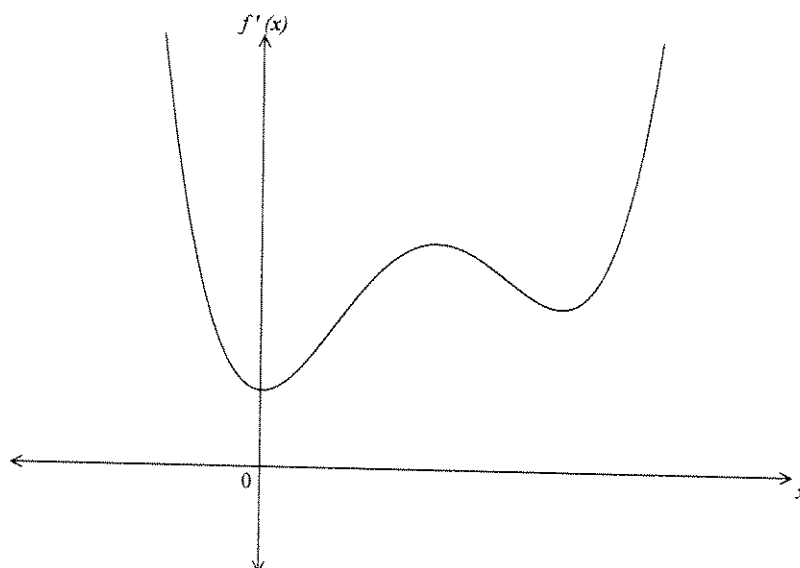
- (b) (i) Sketch the curve $y = 2 \sin \pi x$, $0 \leq x \leq 2$

2

(ii) Find the area bounded by the curve $y = 2 \sin \pi x$ and the x axis from $x = 0$ to $x = 2$

3

(c)



The diagram shows a sketch of the gradient function of the curve $y = f(x)$.

How many stationary points are on the curve $y = f(x)$?

1

End of Question 5

(a) $x^2 - 2Ax + B = 0$ has two different real roots α, β

(i) Show that $A^2 > B$

2

(ii) Find the range of values of B if the sum of the roots is equal to the product of the roots.

3

(b) (i) State the domain of the function $f(x) = \frac{1}{1 + \sqrt{x}}$

1

(ii) Without using calculus, find the range of the function $f(x) = \frac{1}{1 + \sqrt{x}}$

1

(iii) Use Simpson's rule with three function values to give a two decimal place approximation to

$$\int_0^1 \frac{1}{1 + \sqrt{x}} dx$$

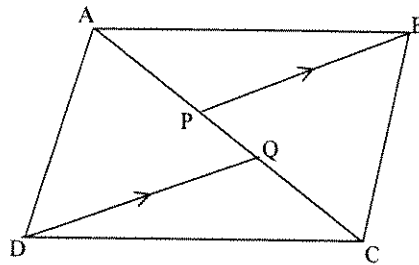
3

(c) Find the focus of the parabola $(x - 1)^2 = 2\left(y + \frac{1}{2}\right)$

2

End of Question 6

(a)



In the diagram, ABCD is a parallelogram.

BP and DQ meet the diagonal AC at P and Q, respectively, where $BP \parallel DQ$

- (i) Explain why $\angle BPQ = \angle DQP$ 1
 - (ii) Prove that $\triangle ABP$ is congruent to $\triangle DCQ$ 3
 - (iii) Deduce that $BQ = DP$ 2
- (b) The population P of a town is known to be changing exponentially.
i.e. $P = P_0 e^{kt}$, P_0, k constants, t time ≥ 0
- (i) Show that $P = P_0 e^{kt}$ satisfies the equation $\frac{dP}{dt} = kP$ 1
 - (ii) At the start of 2001 the population was 25 000 and at the start of 2007 it was 30 000. Prove that the continuous growth rate is approximately 3% p.a. 3
- (c) Simplify $(\sqrt{5} - 2)^4 (\sqrt{5} + 2)^5$ 2

End of Question 7

(a) (i) Show that $\frac{d}{dx}(xe^{-x}) = e^{-x} - xe^{-x}$ **1**

(ii) Hence prove that $\int_0^1 xe^{-x} dx = 1 - 2e^{-1}$ **2**

(b) The region bounded by the curve $y = x + e^{-x}$ and the x axis from $x = 0$ to $x = 1$ is revolved about the x axis.

Prove that the volume of the solid generated is $\frac{\pi}{6}(17 - 24e^{-1} - 3e^{-2})$ **4**

(c) $A(-3, 0)$ and $B(6, 0)$ are two points in the number plane. $P(x, y)$ is any point in the plane such that P is twice as far from B as it is from A . i.e. $PB = 2PA$.

(i) Prove that the cartesian equation of the locus of $P(x, y)$ is $x^2 + 12x + y^2 = 0$ **3**

(ii) Describe the locus of P in precise geometrical terms. **2**

End of Question 8

- (a) A particle is moving on the x axis. Its position at time t seconds is given by

$$x = t^2(t - 6), \quad t \geq 0$$

- | | |
|---|---|
| (i) At what times is the particle at the origin? | 1 |
| (ii) Find expressions for the velocity $\dot{x}$ and the acceleration $\ddot{x}$ | 2 |
| (iii) In what direction is the particle moving at $t = 2$? | 1 |
| (iv) For what values of t is the velocity increasing? | 1 |
| (v) Find the total distance travelled during the first six seconds of the motion. | 2 |

- (b) A rainwater tank is initially empty. The rate, R L/s, at which water is entering the tank is given by

$$R = 1 - \frac{1}{\sqrt{2t + 1}}, \quad t \geq 0 \text{ is the time in seconds}$$

- | | |
|---|---|
| (i) Find the rate at which the tank is filling after one minute. | 1 |
| (ii) The tank is full to its capacity after 66 minutes. Explain why the capacity of the tank is less than 3960 L. | 1 |
| (iii) Determine the capacity of the tank. | 3 |

End of Question 9

(a) Sketch the curve $y = \ln(1 - 2x)$ 2

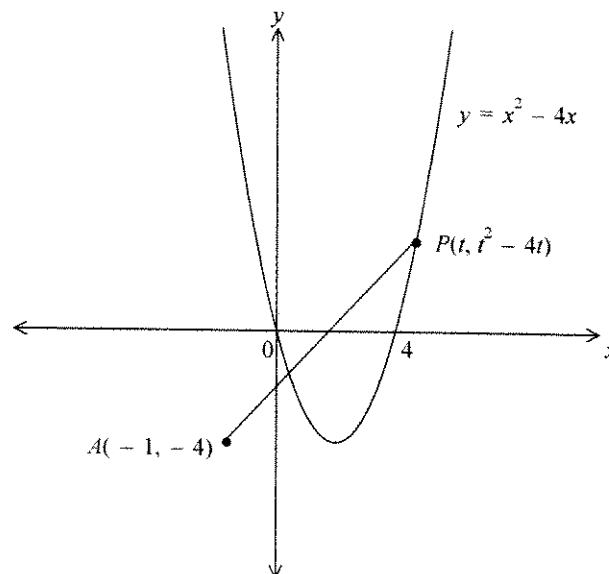
(b) $m \sin x = n \cos x$ where $m, n > 0$ and $0 < x < \frac{\pi}{2}$

Prove that $\sin x \cos x = \frac{mn}{m^2 + n^2}$ 3

(c) Let $f(t) = 2(t-2)^3 + t + 1$ where $f(1) = 0$

By considering $f'(t)$ or otherwise deduce that $f(t) = 0$ only if $t = 1$ 2

(d)



The sketch shows the parabola $y = x^2 - 4x$ and the point $A(-1, -4)$

Let $P(t, t^2 - 4t)$ be any point on the parabola and let $AP^2 = l$

(i) Show that $l = (t+1)^2 + (t-2)^4$ 2

(ii) Hence or otherwise find the minimum length of AP . 3

End of Examination

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note: $\ln x = \log_e x$, $x > 0$

Qn 1

$$(a) \quad 29.1 \quad [\text{PAY } 1 \text{ for } 0.00366]$$

$$(b) \quad = \frac{x(x+1)}{x+1} = x$$

$$(c) \quad -5 \leq 3x+4 \leq 5$$

$$\therefore -9 \leq 3x \leq 1 \Rightarrow -3 \leq x \leq \frac{1}{3}$$

$$(d) \quad \ln 3^x = \ln 2$$

$$\therefore x \ln 3 = \ln 2 \Rightarrow x = \frac{\ln 2}{\ln 3} = 0.63, 2 \text{ d.p.}$$

$$(e) \quad y = 1 + x^{-1}$$

$$\therefore \frac{dy}{dx} = -x^{-2} = -\frac{1}{x^2}$$

$$(f) \quad \frac{(2x+9)^4}{4 \times 2} = \frac{(2x+9)^4}{8}$$

Question 2

$$(a) \therefore 0 = 2x^2 - 5x + 2$$

$$\Rightarrow (2x - 1)(x - 2) = 0$$

$$\begin{cases} x = \frac{1}{2} \text{ or } 2 \\ y = 2 \text{ or } 8 \end{cases} \quad \text{i.e. } (\frac{1}{2}, 2) \text{ and } (2, 8)$$

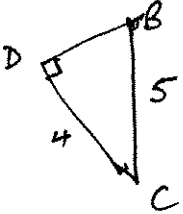
$$(b) (i) \text{ grad } AB = \frac{6}{8} = \frac{3}{4}$$

$$(ii) AB \text{ is } y - 0 = \frac{3}{4}(x + 4)$$

$$\therefore 4y = 3x + 12$$

$$\text{i.e. } 3x - 4y + 12 = 0$$

$$(iii) CD = \frac{12 - 4 + 12}{\sqrt{3^2 + 4^2}} = \frac{20}{5} = 4$$

(iv) we have  $\therefore DB = 3$

$$(c) y' = 3x^2 - 4x = 3 + 4 \text{ at } x = -1 \\ = 7$$

$$\therefore \text{tangent is } y + 3 = 7(x + 1) \text{ will do}$$

$$\text{i.e. } y = 7x + 4$$

Question 3

$$(a) \quad I = 6 \int_0^2 \frac{2x}{x^2+2} dx$$

$$= 6 [\ln(x^2+2)]_0^2 = 6 (\ln 6 - \ln 2) = 6 \ln 3$$

(b) (i) $\angle BPA = \angle DPC$, vertically opposite

$$\frac{PB}{PC} = \frac{PA}{PD} = \frac{2}{3}$$

$\therefore \triangle ABP \parallel \triangle CDP$, sim. Δ test with included angle

(ii) $\angle A = \angle D$, Δ s similar

But these are alternate angles

$\therefore AB \parallel CD$

$$(c) (i) \quad \cos \theta = \frac{7^2 + 15^2 - 13^2}{2 \times 7 \times 15} = \frac{105}{210} = \frac{1}{2}$$

$$\therefore \theta = 60^\circ$$

$$(ii) \quad \text{Area} = \frac{1}{2} \cdot 7 \cdot 15 \sin 60^\circ + \frac{1}{2} \cdot 4 \cdot 15 \sin 60^\circ, \angle CAB = \theta$$

$$= \frac{1}{2} \cdot 11 \cdot 15 \sin 60^\circ$$

$$= \frac{1}{2} \cdot 11 \cdot 15 \cdot \frac{\sqrt{3}}{2} = \frac{165\sqrt{3}}{4} \text{ cm}^2$$

Question 4

(a) (i) arc $AB = 1.5r$

$$\therefore 1.5r + 2r = 1.4$$

$$3.5r = 1.4 \Rightarrow r = \frac{1.4}{3.5} = 0.4$$

(ii) Area = $\frac{1}{2} (0.4)^2 (1.5) \text{ cm}^2 = 0.12 \text{ cm}^2$

(iii) $1.5^\circ = 1.5 \times \frac{180}{\pi}^\circ = 86^\circ$, nearest degree

(b) (i) $d = 2000 - 2007 = -7$

(ii) $T_n = 2007 - 7(n-1) = 12$

$$\therefore 7(n-1) = 1995$$

$$n-1 = 285 \quad \text{or } n = 286$$

Yes. 12 is the 286th term

(iii) We need $\frac{n}{2} (4014 - 7(n-1)) > 0$

$$\Rightarrow 4014 - 7(n-1) > 0$$

$$\therefore 7(n-1) < 4014$$

$$n-1 < \frac{4014}{7} = 573.4 \dots$$

$$\text{ie } n < 574.4 \dots$$

$\Rightarrow$ max. no. of terms is 574

(c) $\therefore 79100 \times 1.002^{40} - M \frac{(1.002^{41} - 1)}{1.002 - 1} = 0$

$$\therefore M = \frac{79100 \times 1.002^{40} \times 0.002}{1.002^{41} - 1} = 2007, \text{ nearest integer}$$

Question 5

(a) (i) $f'(x) = 0 \Rightarrow 3x(2-x) = 0$
 $\therefore x = 0, 2$

$$f'(x) = 6x - 3x^2$$

$$f''(x) = 6 - 6x \quad \& \quad f''(0) > 0 \Rightarrow \cup$$

$$\& \quad f''(2) < 0 \Rightarrow \cap$$

$\therefore$ at $x=0$ there's a minimum turning point
& at $x=2$ there's a maximum turning point

(ii) Since $f'(x) = 6x - 3x^2$

$$\text{then } f(x) = \frac{6x^2}{2} - \frac{3x^3}{3} + C = 3x^2 - x^3 + C$$

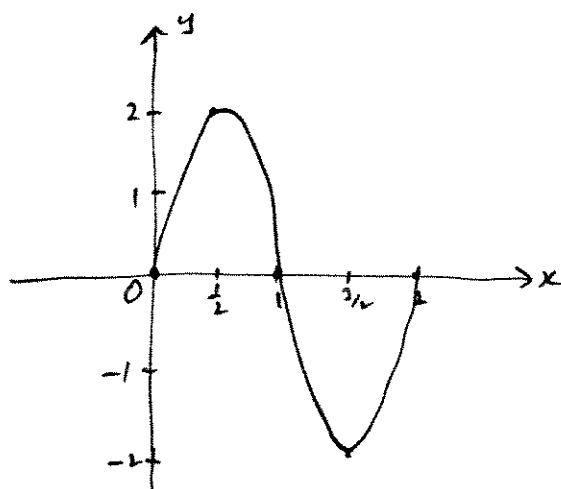
$$\therefore f(2) = 12 - 8 + C = 0 \quad \therefore C = -4$$

$$\therefore f(x) = 3x^2 - x^3 - 4$$

$$\therefore f(-1) = 3 + 1 - 4 = 0$$

(b)

(i)



$$(ii) A = 2 \int_0^1 2 \sin \pi x \, dx = \frac{4}{\pi} [-\cos \pi x]_0^1 \\ = \frac{4}{\pi} (1 + 1) = \frac{8}{\pi} \text{ units}$$

(c) 0 since $f'(x) > 0$ for all x

Question 6

$$(a) (i) \therefore \Delta > 0 \Rightarrow 4A^2 - 4B > 0 \\ \text{or } A^2 > B$$

$$(ii) \text{ We have } 2A = B \\ \therefore 4A^2 = B^2$$

$$\therefore \text{ from (i) , } B^2 > 4B$$

$$\Rightarrow B(B-4) > 0 \Rightarrow B < 0 \text{ or } B > 4$$

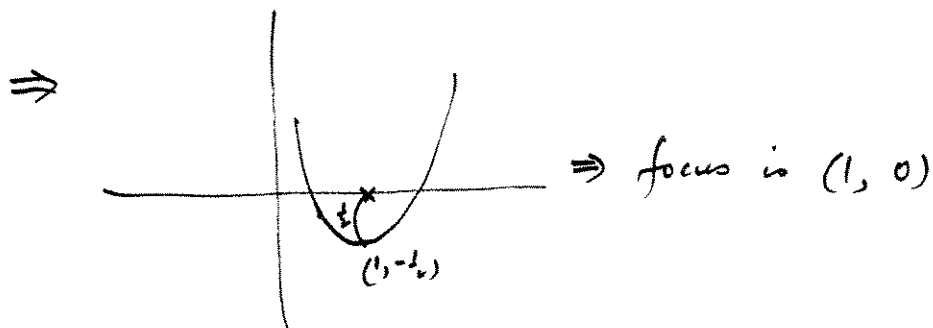
$$(b) (i) x \geq 0$$

$$(ii) \text{ greatest value when } x=0 \text{ giving } y=1 \\ \Rightarrow \text{ range is } 0 < y \leq 1$$

$$(iii) I \approx \frac{1}{6} \cdot 1 \left[1 + \frac{1}{2} + 4 \times \frac{1}{1+\sqrt{0.5}} \right]$$

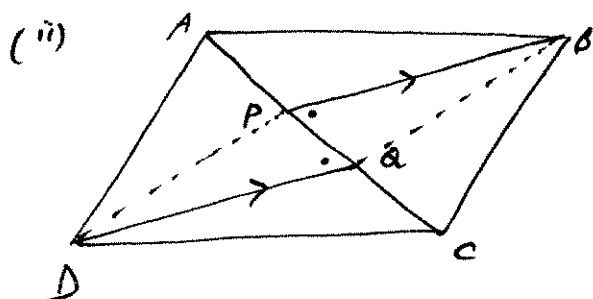
$$= 0.64, \text{ 2 d.p.}$$

$$(c) \text{ Vertex} = (1, -\frac{1}{2}), \quad 4a = 2 \\ a = \frac{1}{2}$$



Question 7

(a) (i) alternate angles in $\parallel$ lines



In Δ s ABP, DCQ

$\angle APB = \angle CQD$, both supplements of equal angles in (i)

$\angle BAP = \angle DCQ$, alt. angles in $\parallel$ lines AB, DC

$AB = DC$, opp. sides of $\parallel$ ogram

$\therefore \Delta ABP \equiv \Delta DCQ$, AAS

(iii) Join BQ, DP

From (ii), $BP = DQ$

and $BP \parallel DQ$, given

$\therefore$ BPDQ is a $\parallel$ ogram [one pair of opp sides equal and parallel]

$\therefore BQ = DP$, opp. sides of $\parallel$ ogram

(b) (i) $\frac{dP}{dt} = P_0 e^{kt} \times k = k(P_0 e^{kt}) = kP$

(ii) 2001, $t=0$, $P=25000$

2007, $t=6$, $P=30000$

$\therefore P = 25000 e^{kt}$

and $30000 = 25000 e^{6k}$

$\therefore e^{6k} = \frac{30}{25} \Rightarrow 6k = \ln\left(\frac{6}{5}\right)$ i.e. $k = \frac{1}{6} \ln\left(\frac{6}{5}\right)$

$\approx 0.0303 \dots$

i.e. growth rate $\approx 3\%$ p.a.

(c) $= (\sqrt{5}-2)(\sqrt{5}+2))^4 (\sqrt{5}+2) = (5-4)^4 (\sqrt{5}+2) = \sqrt{5}+2$

Question 8

$$(a) \quad (i) \quad \frac{d(xe^{-x})}{dx} = e^{-x}(1) + x(-e^{-x}) \\ = e^{-x} - xe^{-x}$$

$$(ii) \quad \text{From (i), } xe^{-x} = e^{-x} - \frac{d(xe^{-x})}{dx}$$

$$\therefore \int_0^1 xe^{-x} dx = \left[-e^{-x} - xe^{-x} \right]_0^1 \\ = -e^{-1} - e^{-1} - (-1 - 0) \\ = 1 - 2e^{-1}$$

$$(b) \quad V = \pi \int_0^1 (x + e^{-x})^2 dx \\ = \pi \int_0^1 x^2 + 2xe^{-x} + e^{-2x} dx \\ = \pi \int_0^1 x^2 + e^{-2x} dx + 2\pi(1 - 2e^{-1}) \quad \text{from (a)(ii)} \\ = \pi \left[\frac{x^3}{3} - \frac{1}{2} e^{-2x} \right]_0^1 + 2\pi(1 - 2e^{-1}) \\ = \pi \left(\frac{1}{3} - \frac{1}{2} e^{-2} - (0 - \frac{1}{2}) \right) + 2\pi(1 - 2e^{-1}) \\ = \pi \left(\frac{5}{6} - \frac{1}{2} e^{-2} \right) + 2\pi(1 - 2e^{-1}) \\ = \frac{\pi}{6} (5 - 3e^{-2} + 12 - 24e^{-1}) \\ = \frac{\pi}{6} (17 - 24e^{-1} - 3e^{-2})$$

$$\therefore (c) \quad (i) \quad PB = 2 PA \quad \therefore PB^2 = 4 \times PA^2$$

$$\Rightarrow (x-6)^2 + y^2 = 4((x+3)^2 + y^2)$$

$$x^2 - 12x + 36 + y^2 = 4x^2 + 24x + 36 + 4y^2$$

$$\text{or} \quad 3x^2 + 3y^2 + 36x = 0$$

$$\text{i.e.} \quad x^2 + 12x + y^2 = 0$$

$$(ii) \text{ Locus is } (x+6)^2 + y^2 = 36$$

i.e. a circle, centre $(-6, 0)$, radius 6

Question 9

(a) (i) $x=0 \Rightarrow t^2(t-6)=0$ i.e. at $t=0, 6$

(ii) $x = t^3 - 6t^2$

$\therefore \ddot{x} = 3t^2 - 12t$ and $\ddot{x} = 6t - 12$

(iii) $t=2, \ddot{x} = 12 - 24 < 0$

$\therefore$ moving from right to left
i.e. negative direction

(iv) v is increasing when $\ddot{x} > 0$

$\Rightarrow 6t - 12 > 0$ or $t > 2$

(v) $v=0, 3t(t-4)=0, t=0, 4$

when $t=4, x = 16(-2) = -32$

$[t=0, x=0; t=6, x=0]$

$\therefore$ distance travelled $= 32 \times 2 = 64$

(b) (i) $t=60, R = 1 - \frac{1}{\sqrt{121}} \text{ L/s} = \frac{10}{11} \text{ L/s}$

(ii) Clearly $R < 1$ for all t

$\therefore$ Capacity $< 66 \times 60 \text{ L} = 3960 \text{ L}$

(iii) $\frac{dV}{dt} = 1 - (2t+1)^{-\frac{1}{2}}$

$\therefore V = t - 2 \frac{(2t+1)^{\frac{1}{2}}}{2} + C$ i.e. $V = t - \sqrt{2t+1} + C$ $[t=0, V=0]$
 $\therefore 0 = 0 - 1 + C, C=1$

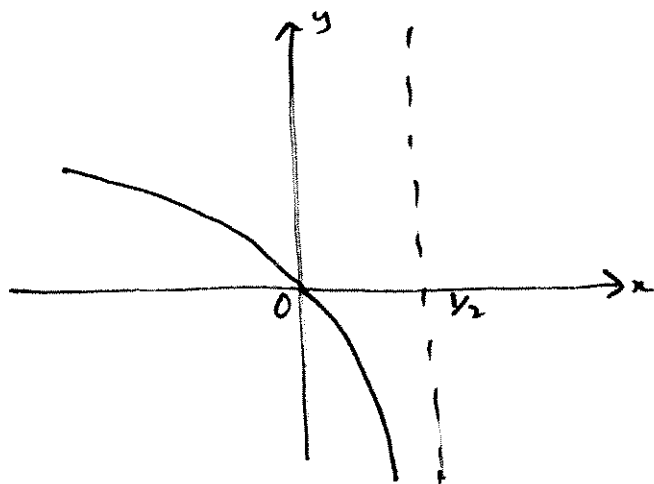
$\therefore V = t - \sqrt{2t+1} + 1$ $[t=66 \text{ min, it's full}]$

$\therefore$ capacity $= 66 \times 60 - \sqrt{793} + 1 = 3872 \text{ L}$

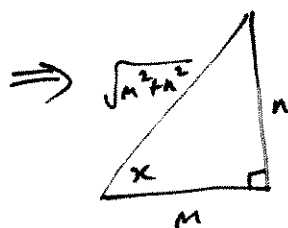
Question 10

(a) $1 - 2x > 0 \Rightarrow 2x < 1 \quad \text{or} \quad x < \frac{1}{2}$

$z = 0, y = 0$



(b) $\therefore m \tan x = n \quad \text{or} \quad \tan x = \frac{n}{m}$



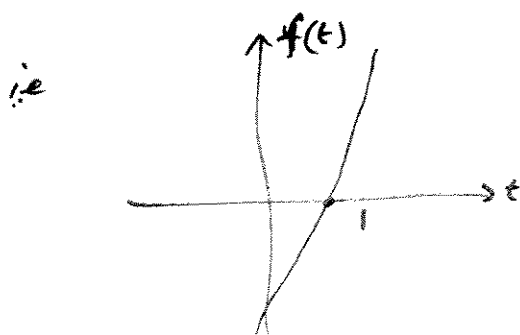
$\therefore \sin x \cos x = \frac{n}{\sqrt{m^2 + n^2}} \cdot \frac{m}{\sqrt{m^2 + n^2}} = \frac{mn}{m^2 + n^2}$

(c) $f'(t) = 6(t-2)^2 + 1 \geq 1 > 0$ for all t

$\therefore f(t)$ increases for all t

$\Rightarrow$ curve could only cut t axis once

$\Rightarrow f(t) = 0$ only if $t = 1$



$$\begin{aligned}
 (d) \quad (i) \quad l &= AP^2 = (t+1)^2 + (t^2 - 4t + 4)^2 \\
 &= (t+1)^2 + ((t-2)^2)^2 \\
 &= (t+1)^2 + (t-2)^4
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \frac{dl}{dt} &= 2(t+1) + 4(t-2)^3 \\
 &= 2(2(t-2)^3 + t+1) \\
 &= 0 \quad \text{only if } t=1 \text{ from (c)}
 \end{aligned}$$

$$\begin{aligned}
 \text{OR USE *} \quad \frac{d^2l}{dt^2} &= 2(6(t-2)^2 + 1) > 0 \text{ if } t=1 \\
 &\quad [\text{in fact } > 0 \text{ for all } t]
 \end{aligned}$$

$$\Rightarrow \min l \text{ when } t=1$$

$$\Rightarrow \min AP \text{ when } t=1 \text{ since } l=AP^2$$

$$\therefore \min \text{ length } AB = \sqrt{2^2 + 1} = \sqrt{5}$$